

Sample Car Data

Single Document

```
{
  "maker": "Tata",
  "model": "Nexon",
  "fuel_type": "Petrol",
  "transmission": "Automatic",
  "engine": {
    "type": "Turbocharged",
    "cc": 1199,
    "torque": "170 Nm"
  },
  "features": [
    "Touchscreen",
    "Reverse Camera",
    "Bluetooth Connectivity"
  ],
  "sunroof": false,
  "airbags": 2
}
```

Many Document [{}, {}, {}]

```
[
  {
    "maker": "Hyundai",
    "model": "Creta",
    "fuel_type": "Diesel",
    "transmission": "Manual",
    "engine": {
      "type": "Naturally Aspirated",
      "cc": 1493,
      "torque": "250 Nm"
    },
    "features": [
      "Sunroof",
      "Leather Seats",
      "Wireless Charging",
      "Ventilated Seats",
      "Bluetooth"
    ]
  },
  {
    "maker": "Ford",
    "model": "Mustang",
    "fuel_type": "Petrol",
    "transmission": "Automatic",
    "engine": {
      "type": "Turbocharged",
      "cc": 2000,
      "torque": "350 Nm"
    },
    "features": [
      "Sunroof",
      "Leather Seats",
      "Wireless Charging",
      "Ventilated Seats",
      "Bluetooth"
    ]
  },
  {
    "maker": "BMW",
    "model": "X5",
    "fuel_type": "Petrol",
    "transmission": "Automatic",
    "engine": {
      "type": "Turbocharged",
      "cc": 3000,
      "torque": "450 Nm"
    },
    "features": [
      "Sunroof",
      "Leather Seats",
      "Wireless Charging",
      "Ventilated Seats",
      "Bluetooth"
    ]
  }
]
```

```
],
"sunroof": true,
"airbags": 6
},
{
"maker": "Maruti Suzuki",
"model": "Baleno",
"fuel_type": "Petrol",
"transmission": "Automatic",
"engine": {
"type": "Naturally Aspirated",
"cc": 1197,
"torque": "113 Nm"
},
"features": [
"Projector Headlamps",
"Apple CarPlay",
"ABS"
],
"sunroof": false,
"airbags": 2
},
{
"maker": "Mahindra",
"model": "XUV500",
"fuel_type": "Diesel",
"transmission": "Manual",
"engine": {
"type": "Turbocharged",
"cc": 2179,
"torque": "360 Nm"
},
"features": [
"All-Wheel Drive",
"Navigation System",
"Cruise Control"
],
"sunroof": true,
"airbags": 6
},
{
"maker": "Honda",
"model": "City",
"fuel_type": "Petrol",
"sunroof": true,
"airbags": 6
}
```

```
"transmission": "Automatic",
"engine": {
  "type": "Naturally Aspirated",
  "cc": 1498,
  "torque": "145 Nm"
},
"features": [
  "Keyless Entry",
  "Auto AC",
  "Multi-angle Rearview Camera"
],
"sunroof": false,
"airbags": 4
}
]
```

Data for Relationship

users collection

```
[
  {
    "_id": "user1",
    "name": "Amit Sharma",
    "email": "amit.sharma@example.com",
    "phone": "+91-987654210",
    "address": "MG Road, Mumbai, Maharashtra"
  },
  {
    "_id": "user2",
    "name": "Priya Verma",
    "email": "priya.verma@example.com",
    "phone": "+91-987654211",
    "address": "Nehru Place, New Delhi, Delhi"
  },
  {
    "_id": "user3",
    "name": "Rahul Singh",
    "email": "rahul.singh@example.com",
    "phone": "+91-987654212",
    "address": "Sector 18, Noida, Uttar Pradesh"
  },
  ,
]
```

```
},
{
  "_id": "user4",
  "name": "Anjali Nair",
  "email": "anjali.nair@example.com",
  "phone": "+91-987654213",
  "address": "Marine Drive, Kochi, Kerala"
},
{
  "_id": "user5",
  "name": "Vikram Desai",
  "email": "vikram.desai@example.com",
  "phone": "+91-987654214",
  "address": "Park Street, Kolkata, West Bengal"
}
]
```

orders collection

```
[
  {
    "_id": "order1",
    "user_id": "user1",
    "product": "Laptop",
    "amount": 50000,
    "order_date": "2024-08-01"
  },
  {
    "_id": "order2",
    "user_id": "user2",
    "product": "Mobile Phone",
    "amount": 15000,
    "order_date": "2024-08-05"
  },
  {
    "_id": "order3",
    "user_id": "user1",
    "product": "Headphones",
    "amount": 2000,
    "order_date": "2024-08-10"
  },
  {
    "_id": "order4",
    "user_id": "user3",
    "product": "Smartwatch",
    "amount": 3000,
    "order_date": "2024-08-15"
  }
]
```

```

      "user_id": "user3",
      "product": "Tablet",
      "amount": 25000,
      "order_date": "2024-08-12"
    },
    {
      "_id": "order5",
      "user_id": "user4",
      "product": "Smart Watch",
      "amount": 8000,
      "order_date": "2024-08-15"
    }
  ]

```

Data for Aggregate Framework

```

[{"maker": "Hyundai", "model": "Creta", "fuel_type": "Diesel", "transmission": "Manual", "engine": {"type": "Naturally Aspirated", "cc": 1493, "torque": "250 Nm"}, "features": ["Sunroof", "Leather Seats", "Wireless Charging", "Ventilated Seats", "Bluetooth"], "sunroof": true, "airbags": 6, "price": 1500000, "owners": [{"name": "Raju", "purchase_date": "2021-03-15", "location": "Mumbai"}, {"name": "Shyam", "purchase_date": "2023-01-10", "location": "Delhi"}], "service_history": [{"date": "2022-04-10", "service_type": "Oil Change", "cost": 5000}, {"date": "2023-07-18", "service_type": "Brake Replacement", "cost": 12000}], {"maker": "Maruti Suzuki", "model": "Baleno", "fuel_type": "Petrol", "transmission": "Automatic", "engine": {"type": "Naturally Aspirated", "cc": 1197, "torque": "113 Nm"}, "features": ["Projector Headlamps", "Apple CarPlay", "ABS"], "sunroof": false, "airbags": 2, "price": 800000, "owners": [{"name": "Baburao", "purchase_date": "2020-08-22", "location": "Pune"}], "service_history": [{"date": "2021-05-12", "service_type": "Tire Rotation", "cost": 2000}, {"date": "2022-11-05", "service_type": "Battery Replacement", "cost": 7000}], {"maker": "Mahindra", "model": "XUV500", "fuel_type": "Diesel", "transmission": "Manual", "engine": {"type": "Turbocharged", "cc": 2179, "torque": "360 Nm"}, "features": ["All-Wheel Drive", "Navigation System", "Cruise Control"], "sunroof": true, "airbags": 6, "price": 1800000, "owners": [{"name": "Raju", "purchase_date": "2019-11-30", "location": "Bangalore"}, {"name": "Shyam", "purchase_date": "2022-02-15", "location": "Hyderabad"}], "service_history": [{"date": "2021-02-25", "service_type": "Transmission Repair", "cost": 35000}, {"date": "2023-03-10", "service_type": "Tire Replacement", "cost": 15000}], {"maker": "Honda", "model": "City", "fuel_type": "Petrol", "transmission": "Automatic", "engine": {"type": "Naturally Aspirated", "cc": 1498, "torque": "145 Nm"}, "features": ["Keyless Entry", "Auto AC", "Multi-angle Rearview Camera"], "sunroof": false, "airbags": 4, "price": 1200000, "owners": [{"name": "Baburao", "purchase_date": "2020-05-20", "location": "Chennai"}], "service_history": [{"date": "2021-08-10", "service_type": "Oil Change", "cost": 5000}, {"date": "2022-10-25", "service_type": "Brake Replacement", "cost": 10000}], {"maker": "Tata", "model": "Nexon", "fuel_type": "Petrol", "transmission": "Automatic", "engine":

```

```
{"type":"Turbocharged","cc":1199,"torque":"170 Nm"},"features":["Touchscreen","Reverse Camera","Bluetooth Connectivity"],"sunroof":false,"airbags":2,"price":1100000,"owners":[{"name":"Raju","purchase_date":"2021-12-05","location":"Kolkata"}],"service_history":[{"date":"2022-12-01","service_type":"Oil Change","cost":6000},{date":"2023-06-15","service_type":"Tire Rotation","cost":3000}]}, {"maker":"Hyundai","model":"Venue","fuel_type":"Petrol","transmission":"Automatic","engine":{"type":"Turbocharged","cc":998,"torque":"172 Nm"},"features":["Sunroof","Touchscreen Infotainment","Keyless Entry","Rear Camera","Cruise Control"],"sunroof":true,"airbags":4,"price":1200000,"owners":[{"name":"Amit","purchase_date":"2020-05-20","location":"Bangalore"}, {"name":"Priya","purchase_date":"2022-11-05","location":"Chennai"}],"service_history":[{"date":"2021-07-15","service_type":"Oil Change","cost":4000},{date":"2023-03-22","service_type":"Tire Replacement","cost":8000}]}, {"maker":"Hyundai","model":"i20","fuel_type":"Petrol","transmission":"Manual","engine":{"type":"Naturally Aspirated","cc":1197,"torque":"114 Nm"},"features":["Apple CarPlay","ABS","Projector Headlamps","Wireless Charging"],"sunroof":false,"airbags":2,"price":900000,"owners":[{"name":"Rohit","purchase_date":"2021-06-15","location":"Delhi"}],"service_history":[{"date":"2022-09-10","service_type":"Battery Replacement","cost":7000},{date":"2023-05-25","service_type":"Tire Rotation","cost":2500}]}, {"maker":"Maruti Suzuki","model":"Swift","fuel_type":"Petrol","transmission":"Manual","engine":{"type":"Naturally Aspirated","cc":1198,"torque":"113 Nm"},"features":["Touchscreen Infotainment","ABS","Keyless Entry","Rear Parking Sensors"],"sunroof":false,"airbags":2,"price":750000,"owners":[{"name":"Vijay","purchase_date":"2019-03-20","location":"Hyderabad"}],"service_history":[{"date":"2020-05-18","service_type":"Oil Change","cost":3000},{date":"2022-08-10","service_type":"Brake Replacement","cost":5000}]}, {"maker":"Tata","model":"Harrier","fuel_type":"Diesel","transmission":"Automatic","engine":{"type":"Turbocharged","cc":1956,"torque":"350 Nm"},"features":["Panoramic Sunroof","Leather Upholstery","Terrain Response System","Auto-Dimming IRVM"],"sunroof":true,"airbags":6,"price":2000000,"owners":[{"name":"Deepak","purchase_date":"2022-01-10","location":"Mumbai"}],"service_history":[{"date":"2022-10-15","service_type":"Transmission Repair","cost":45000},{date":"2023-04-20","service_type":"Brake Replacement","cost":15000}]}, {"maker":"Honda","model":"Amaze","fuel_type":"Diesel","transmission":"Manual","engine":{"type":"Naturally Aspirated","cc":1498,"torque":"200 Nm"},"features":["Keyless Entry","Auto AC","Rear Parking Camera","Cruise Control"],"sunroof":false,"airbags":4,"price":1000000,"owners":[{"name":"Anil","purchase_date":"2020-11-25","location":"Kolkata"}],"service_history":[{"date":"2021-12-10","service_type":"Oil Change","cost":4500},{date":"2022-08-15","service_type":"Tire Rotation","cost":2500}]}, {"maker":"Tata","model":"Nexon EV","fuel_type":"Electric","transmission":"Automatic","engine":{"type":"Electric Motor","battery_capacity":"30.2 kWh","torque":"245 Nm"},"features":["Touchscreen Infotainment","Wireless Charging","Connected Car Tech","Sunroof"],"sunroof":true,"airbags":6,"price":1400000,"owners":[{"name":"Vikas","purchase_date":"2021-05-20","location":"Bangalore"}],"service_history":[{"date":"2022-06-10","service_type":"Battery Check","cost":0},{date":"2023-03-15","service_type":"Tire Rotation","cost":3000}]}, {"maker":"Hyundai","model":"Kona
```

```
Electric","fuel_type":"Electric","transmission":"Automatic","engine":{"type":"Electric
Motor","battery_capacity":"39.2 kWh","torque":"395 Nm"},"features":["Wireless Charging","Ventilated
Seats","Sunroof","Auto AC"],"sunroof":true,"airbags":6,"price":2300000,"owners":
[{"name":"Sneha","purchase_date":"2022-01-15","location":"Mumbai"}],"service_history":
[{"date":"2022-09-10","service_type":"Battery Check","cost":0}, {"date":"2023-06-
05","service_type":"Brake Replacement","cost":8000}]], {"maker":"Maruti
Suzuki","model":"WagonR","fuel_type":"CNG","transmission":"Manual","engine":{"type":"Naturally
Aspirated","cc":998,"torque":"90 Nm"},"features":["Manual AC","ABS","Power
Windows"],"sunroof":false,"airbags":2,"price":600000,"owners":
[{"name":"Rahul","purchase_date":"2019-07-22","location":"Delhi"}],"service_history":[{"date":"2020-
11-10","service_type":"CNG Kit Checkup","cost":2000}, {"date":"2021-08-15","service_type":"Tire
Rotation","cost":1500}]],
{"maker":"Honda","model":"Amaze","fuel_type":"CNG","transmission":"Manual","engine":
{"type":"Naturally Aspirated","cc":1199,"torque":"110 Nm"},"features":["Keyless Entry","Auto
AC","Rear Parking Camera","Cruise Control"],"sunroof":false,"airbags":4,"price":800000,"owners":
[{"name":"Sanjay","purchase_date":"2021-03-18","location":"Pune"}],"service_history":[{"date":"2021-
09-10","service_type":"CNG Kit Checkup","cost":2500}, {"date":"2022-05-15","service_type":"Oil
Change","cost":3500}]]]
```

Python Program

```
from pymongo import MongoClient

# Replace with your MongoDB connection string
client = MongoClient('mongodb://localhost:27017/')

db = client.todo_db # Create or connect to a database named "todo_db"
tasks_collection = db.tasks # Create or connect to a collection named "tasks"

def create_task(description):
    task = {
        'description': description
    }
    result = tasks_collection.insert_one(task)
    print(f'Task created with id: {result.inserted_id}')

def read_tasks():
```

```

tasks = tasks_collection.find()
print("\nYour To-Do List:")
for task in tasks:
    print(f"- {task['description']}")

while True:
    print("\n1. Create Task")
    print("2. View Tasks")
    print("3. Exit")

    choice = input("Enter your choice: ")

    if choice == '1':
        description = input("Enter task description: ")
        create_task(description)
    elif choice == '2':
        read_tasks()
    elif choice == '3':
        break
    else:
        print("Invalid choice. Please try again.")

```

Queries Used in Video

```

db.cars.aggregate([
    {$group: {
        _id: "$fuel_type",
        TotalCars: {$sum: 1
        }
    }
    }
])

```

```

db.cars.aggregate([
    {$match: {
        maker: "Hyundai"
    }
    },
    {$project: {
        _id: 0,
        maker: 1,
        model: 1,

```



```

        fuel_type: 1
    }
},
{$sort: {model: 1
    }
}
])

db.cars.aggregate([
    {$sortByCount: "$maker"
    }
])

db.cars.aggregate([
    {$match: {maker: "Hyundai"
    }
},
{$project: {
    _id: 0,
    Car_Name: {$toUpper: {$concat: [
        "$maker",
        " ",
        "$model"
    ]
    }
    }
},
{$out: "hyundai_cars"
}
])

db.cars.aggregate([
    {$project: {
    _id: 0,
    model: 1,
    is_diesel: {
        $regexMatch: {
            input: "$fuel_type",
            regex: "Dies"
        }
    }
    }
},
{$out: "diesel_cars"
}
])

```

```

db.cars.aggregate([
  {$project: {
    _id: 0,
    model: 1,
    price: 1
  }
},
{$addFields: {
  price_in_lakhs: {
    $round: [
      {$divide: [
        "$price",
        90000
      ]
    },
    1
  ]
}
}
])

```

```

db.cars.aggregate([
  { $match: { maker: "Hyundai"
  }
},
{ $set: {
  total_service_cost: {
    $sum: "$service_history.cost"
  }
}
},
{ $project: {
  _id: 0,
  model: 1,
  total_service_cost: 1
}
}
])

```

```

db.cars.aggregate([
  {$project: {
    _id: 0,
    maker: 1,
    model: 1,

```

```

fuel_cat: {
  $cond: {
    if: {$eq: [
      "$fuel_type",
      "Petrol"
    ]
    },
    then: "Petrol Car",
    else: "Non_petrol Car"
  }
}
})

```

```

db.cars.aggregate([
  {$project: {
    _id: 0,
    maker: 1,
    model: 1,
    budget_cat: {
      $switch: {
        branches: [
          { //If car is less than 5 Lakhs
            case: {$lt: [
              "$price",
              500000
            ]
            },
            then: "Budget"
          },
          {
            //Car price range between 5-10 lakhs
            case: {$and: [
              {$gte: [
                "$price",
                500000
              ]
              },
              {$lt: [
                "$price",
                1000000
              ]
              }
            ]
          }
        ]
      }
    }
  }
})

```

```

        },
        then: "Mid-Range"
    },
    {
        case: {$gte: [
            "$price",
            1000000
        ]
        },
        then: "Premium"
    }
],
default: "Unknown"
}
}
}
})

```

```

db.cars.aggregate(
    {$project:
        {
            _id: 0,
            model: 1,
            date: "$$NOW"
        }
    })

```

```

db.users.aggregate([
    {$lookup:{
        from:"orders",
        localField:"_id",
        foreignField:"user_id",
        as: "orders"
    }}
])

```

```

db.createCollection("users3", {
    validator: {
        $jsonSchema: {
            bsonType: "object",
            required: ["name", "phone"],
            properties:{
                name:{
                    bsonType: "string",
                    description: "Name should be string"
                }
            }
        }
    }
})

```

```
    }  
  }  
}  
},  
validationLevel: "strict",  
validationAction: "error"  
})  
  
db.users.find().count()  
db.users.find({name: 'Anthony Hurst'})  
db.movies.find({title: 'The Ace of Hearts'}).explain("executionStats")  
db.users.createIndex({ name: 1 })  
db.users.dropIndex("name")
```